SIGMOID | Bike-Sharing Case Study

Demand forecasting, operational decision support, and fleet-sizing assumptions | Dataset: Capital Bikeshare, Washington D.C., 2011–2012

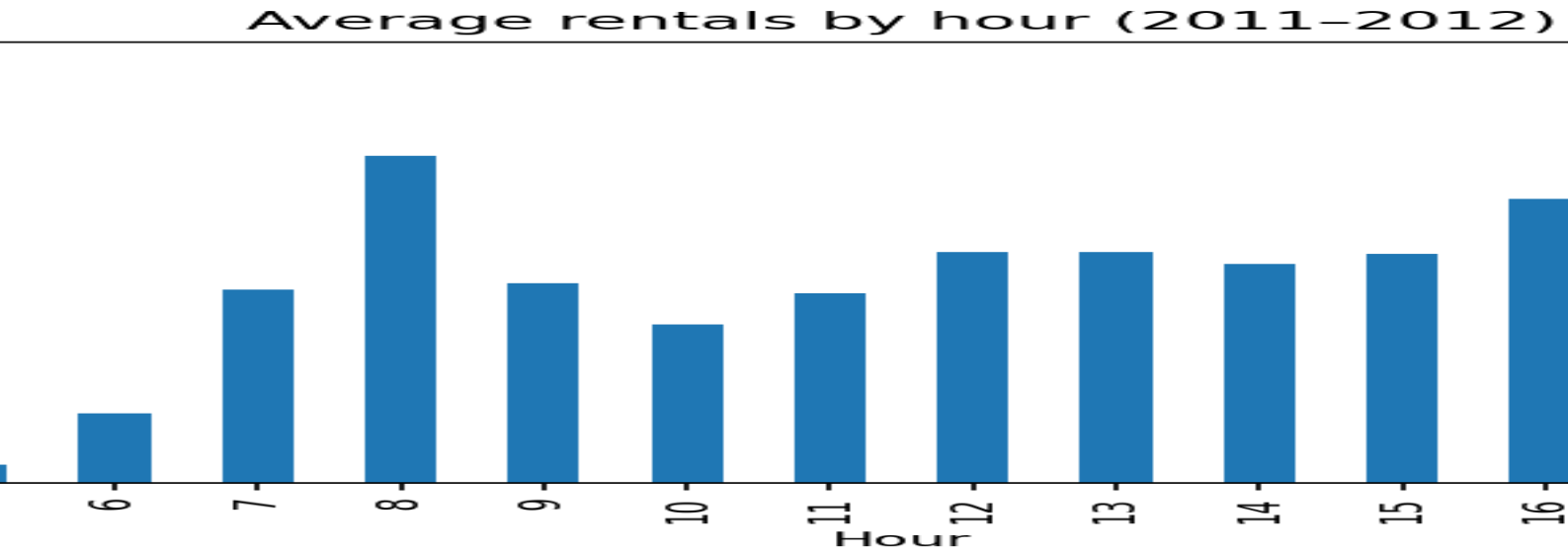
1. Executive recommendation

Do not approve a blanket fleet purchase based on these data alone. Forecast citywide trip-start demand and target operational availability during commuter peaks. Pilot bike rebalancing and instrument unmet demand at station level before deciding whether to expand the fleet. Historical completed rentals cannot reveal rejected rental requests or the exact number of bikes needed.

2. Data quality and understanding

Hourly observations: **17,379**; daily observations: **731**. Date range: 2011-01-01 to 2012-12-31. Null cells: 0 hourly / 0 daily. Duplicate hourly timestamps: 0. Negative rental counts: 0. Every row satisfies casual + registered = cnt; hourly and daily totals reconcile. There are **165** absent hourly timestamps across the full calendar; these were not imputed as zero.

Target: cnt (completed rentals per hour). Inputs: hour, weekday, working-day/holiday flags, season, year/month, weather category, normalized temperature, apparent temperature, humidity, windspeed, and cyclical time features. Exclude casual and registered (direct target leakage), instant (row identifier), and the target itself. The readme documents normalization: temp × 41°C, atemp × 50°C, humidity × 100%, windspeed × 67.



3. Testable hypothesis and experiment

H0: Mean hourly rentals during working-day commute hours (07–09 and 17–19) are no higher than other working-day hours.
H1: Commute-hour mean is higher. This is a prespecified comparison, not an intervention.

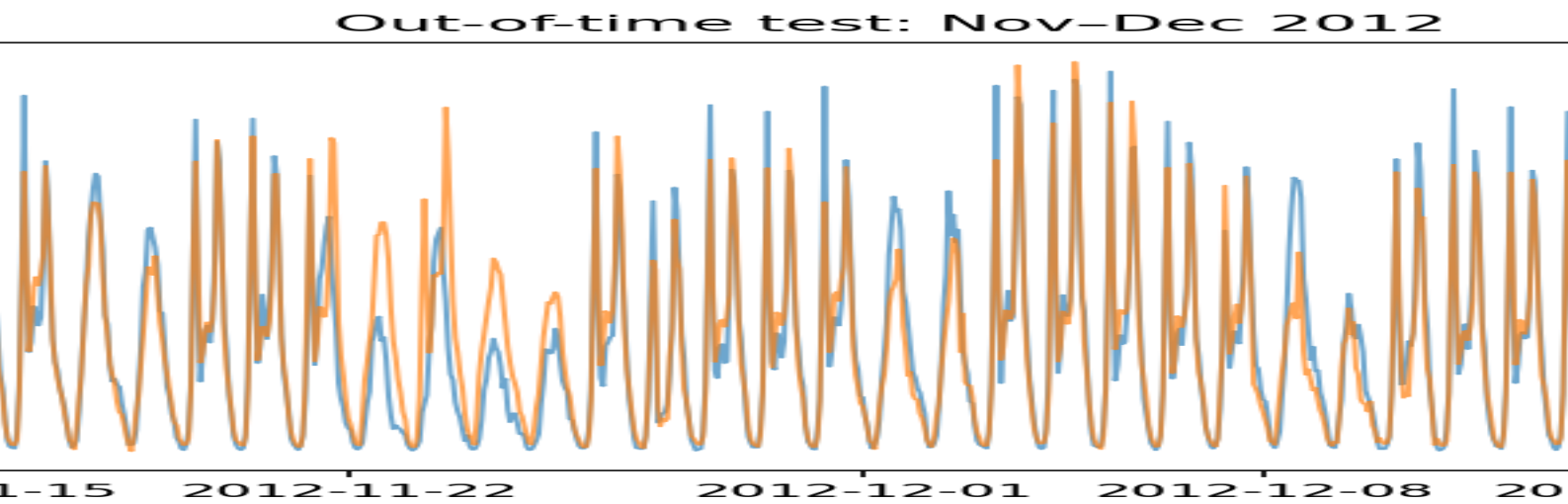
Observed means: **396.0** versus **125.1** rentals/hour, raw difference **270.9**. Welch test p-value is smaller than machine reporting precision (report $p < 0.001$). The more dependence-aware day-block bootstrap yields a 95% CI of **260.1 to 280.5** rentals/hour for the within-day mean difference over 499 working days. This supports association, not causality; season, special events and commuting patterns may confound it.

4. Forecasting experiment

Chronological train: Jan 2011–Sep 2012; validation: Oct 2012; untouched test: Nov–Dec 2012. Compare mean baseline, random forest, and histogram gradient boosting using MAE. Select only on validation MAE; refit selected model on train + validation and evaluate once on test. Weather features in the backtest are observed, not weather forecasts.

Model	Validation MAE	Initial test MAE
mean_baseline	192.67	139.51
random_forest	45.09	50.30
hist_gradient_boosting	42.91	44.46

Selected: **hist_gradient_boosting**. Final refit held-out test MAE **43.23** rentals/hour, RMSE **67.72**, R^2 **0.849**. This is historical backtest performance, not proof of current-year generalization. The 90th percentile absolute validation error is 102.9 rentals/hour, a rough error buffer rather than a calibrated prediction interval.



5. Estimating bikes needed: explicit scenario assumptions

Predicted 95th-percentile hourly trip starts on the test period: **552.0**. Proxy active bikes = hourly trip starts × assumed average trip duration (hours). Proxy fleet = active bikes / assumed fleet utilization. This uses an average-rate approximation and ignores within-hour surges, station distribution, maintenance and rebalancing. It is **not** a purchasing estimate.

Trip duration	Utilization	Active bikes	Illustrative fleet
15 min	70%	138	198
15 min	85%	138	163
30 min	70%	276	395
30 min	85%	276	325
45 min	70%	414	592
45 min	85%	414	488

6. Operations prototype

The supplied Streamlit application accepts natural-language questions about the busiest hour, test-set accuracy, working-day commute effects, fleet assumptions, a historical test date, and whether to buy bikes. It uses transparent intent rules and saved model outputs rather than an external generative API. It explicitly refuses to claim live station availability or exact procurement requirements.

7. Business action and follow-up experiment

Pilot a forecast-guided rebalancing policy in a subset of stations against matched control stations for 4–6 weeks. Record station inventory every 5–15 minutes, attempted but unsuccessful checkouts, trip start/end and duration, bike downtime, transfer costs, and service complaints. Primary metric: failed checkout attempts per 1,000 attempts; guardrails: cost per

successful ride, empty-station minutes, and complaints. Randomize comparable station clusters where feasible, compare pre/post changes against controls, and avoid spillovers from bikes moved between treatment/control areas. Only consider purchasing after measuring residual stockouts despite efficient rebalancing and comparing marginal purchase costs against lost demand.

8. Limitations and reproducibility

This public historical dataset aggregates rental counts at citywide hourly/daily levels. It cannot identify which stations are empty, which bikes are idle, trip durations, unmet requests, current inventory, actual utilization, or purchase costs. Missing time slots may reflect absent records, not zero demand. Model evaluation uses historical weather values; production would require forecasted weather and monitoring for drift. See analysis.py, app.py, outputs/results.json, and README.md for runnable methods and outputs.

Source: Fanaee-T, H. and Gama, J. (2013), “Event labeling combining ensemble detectors and background knowledge,” Progress in Artificial Intelligence. DOI: 10.1007/s13748-013-0040-3.